Risk Estimation of Preeclampsia

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Abstract

Code Availability: https://github.com/AhmedKhan-GH/repnet

Reliably identifying preeclampsia, a leading cause of maternal and fetal morbidity, from routine clinical data remains challenging. Recent work has shown that preeclampsia leaves a detectable signature on the standard 12-lead electrocardiogram (ECG) [8], a cheap, non-invasive, and widely available test. We present *RepNet*, a pair of machine-learning models that aim to identify preeclampsia directly from 12-lead ECGs. RepNet-CNN is a compact, channel-independent convolutional neural network (29,490 parameters) in which all twelve leads pass through a shared-weight 1D backbone and are fused by a single linear layer; RepNet-RF is a decision-tree ensemble trained on interpretable ECG features extracted with NeuroKit2 and GE MUSE-reported measurements. Both models were trained and evaluated on 2,178 clinical 12-lead ECGs (335 preeclamptic, 15.4% prevalence) from 1,383 patients at UC Davis Health, using 30 patient-grouped 80/20 splits to prevent patient-level leakage. RepNet-CNN attains a mean AUROC of 0.709 ± 0.049 (best split 0.779), and RepNet-RF a mean AUROC of 0.757 ± 0.039. Gradient-based saliency and decision-tree feature importance independently converge on clinically established markers (QRS morphology, ST-segment changes, and the T_{pe} interval), showing that both approaches recover known cardiac effects of preeclampsia and that 12-lead ECGs carry discriminative signal for low-cost screening.

1 Introduction

Preeclampsia is a hypertensive disorder of pregnancy and one of the leading causes of maternal and fetal harm. It affects an estimated 2–8% of pregnancies worldwide [37] and with related hypertensive disorders claims more than 76,000 lives a year [8]. Left unmanaged it can progress rapidly to eclampsia, organ failure, and death [12]. Yet routine screening still relies on blood pressure, urine protein, and blood panels that often catch the disorder only once it is clinically apparent [2]. The burden falls hardest on low-resource settings where specialist obstetric care is scarce and a cheap, widely available test would help most.

The standard 12-lead electrocardiogram (ECG) is a compelling candidate for such a test. It is cheap, non-invasive, and already ubiquitous in clinical practice. There is also a clear physiological reason to expect it to carry information about preeclampsia: the disorder imposes measurable cardiac stress that the ECG can capture. Raffaelli et al. found that preeclamptic women exhibit altered ventricular repolarization and increased QT dispersion on standard ECG [25]. A 2025 systematic review and meta-analysis of 57 studies quantified this effect, reporting that third-trimester preeclampsia prolongs the heart-rate-corrected QT (QTc) interval by 21.7 ms relative to normal pregnancy [4]. Beyond repolarization, Angeli et al. report that left atrial abnormality on a standard ECG is associated with a roughly fourfold increase in the risk of developing a hypertensive disorder of pregnancy [5].

To our knowledge, the only prior study to diagnose preeclampsia from ECG waveforms is Butler et al., who trained a modified ResNet convolutional neural network on raw 12-lead ECGs and identified preeclampsia with an AUC of 0.85 on a holdout set and 0.81 on an external cohort [8], building on an earlier demonstration of the same approach [9]. Their discussion points to two future improvements. One reduces the number of leads; the other uses a single-lead (lead I) model that smart wearables could mimic. The first aligns with our own goal of cutting cost and complexity by requiring fewer leads. Their follow-up work supports the second. There a single-lead (lead I) model predicts fatal coronary heart disease, and a clinical ECG and an Apple Watch agree on its risk classification for 99% of participants [10]. Whether a single-lead model could help our task remains open, since several of our features cannot be computed from one lead alone. More broadly, reviews of machine learning for preeclampsia note that most models are built on electronic-health-record features rather than ECG waveforms [37].

Lead reduction is feasible in principle. Xue shows that a four-lead subset (I, II, V2, and V4) fed into a deep neural network reproduces full 12-lead morphology interpretations and detects abnormalities such as bundle branch block, ischemia, and myocardial infarction about as well [32]. Much of the diagnostic information thus survives aggressive lead reduction. Whether this transfers to preeclampsia is unclear, because the diagnosis is not tied to lead-specific morphology in the same way. The result still bears on our setting, because the subset keeps the limb leads I and II that several of our engineered features need, such as the mean electrical axis [1].

Decision-tree models have likewise shown promise for preeclampsia, though always from tabular clinical data rather than the ECG. Bülez et al. trained a LightGBM classifier on electronic health records and found hemoglobin, age, and liver-enzyme levels (AST and ALT) to be the strongest predictors, reaching an AUROC of 0.832 [11]. Zhang et al. instead fit an XGBoost model to early-pregnancy urine metabolomics, isolating seven biomarkers that detect preeclampsia with an AUROC of 0.856 [35]. No such model has yet been built on features drawn from the ECG. Tree ensembles are also far cheaper to run than convolutional networks. They may give up some accuracy, but that efficiency makes them attractive for clinics with limited or aging hardware.

Against this backdrop we develop *RepNet*, a pair of machine-learning models that identify preeclampsia directly from 12-lead ECGs. We train and evaluate both on an independent cohort of 2,178 clinical ECGs from 1,383 patients at UC Davis Health, of which 335 (15.4%) are preeclamptic. The first model, RepNet-CNN, is a compact channel-independent convolutional network of 29,490 parameters. Every lead passes through a single shared-weight 1D backbone, and one linear layer fuses the resulting per-lead features. The second, RepNet-RF, is a decision-tree ensemble trained on interpretable ECG features extracted with NeuroKit2 and GE MUSE-reported measurements. We evaluate both with patient-grouped cross-validation that guarantees no patient appears in both training and test partitions.

2 Dataset

RepNet was developed on a retrospective cohort of clinical 12-lead ECGs from pregnant patients at UC Davis Health. Each recording is a standard 10-second, 12-lead ECG sampled at 500 Hz and comes with the vendor (GE MUSE) interpretation and a de-identified patient identifier. We label a recording preeclamptic when its diagnosis is *preeclampsia or other hypertensive disorder of pregnancy* and normal otherwise. The raw export held 2,483 metadata entries, of which 2,191 had a matching waveform file. We then discarded any recording with a non-finite sample, a flat lead (raw standard deviation $\sigma_\ell < 10^{-4}$), or a missing patient identifier. This removed 13 recordings and two patients and left a clean cohort of 2,178 ECGs from 1,383 patients, of which 335 (15.4%) are preeclamptic. Table 1 reports the cohort before and after this step. The data is protected health information and is available only under IRB approval and a data-use agreement.

Table 1: UCDH cohort before and after removal of broken recordings.

	Before	Removed	After
Patients	1,385	2	1,383
ECG recordings	2,191	13	2,178
Preeclamptic	337	2	335
Normal	1,854	11	1,843
Prevalence	15.4%	–	15.4%

Two properties of this cohort shape the rest of the paper. First, the classes are heavily imbalanced. A trivial model that labels every recording normal is already 84.6% accurate while catching no cases, so we report threshold-free measures (AUROC and AUPRC) rather than accuracy. Second, the 1,383 patients contribute the 2,178 recordings unevenly (Table 2). Most appear once and 428 (31%) appear more than once, with one patient contributing as many as 16 and a mean of 1.6 overall. The label is assigned at the patient level, so no patient carries conflicting labels, but a naive recording-level split would still place the same patient in both training and test data and inflate performance. We therefore group every split by patient (Section 4) so that no patient is seen during both training and evaluation. We use scikit-learn's `StratifiedGroupKFold` [24], which partitions by patient while preserving the class ratio in each fold.

Table 2: Distribution of ECG recordings per patient across the cohort.

Recordings/patient	Patients	Recordings
1	955	955
2	257	514
3	101	303
4–5	47	202
6–10	16	114
11–16	7	90
Total	**1,383**	**2,178**

2.1 Preprocessing

Raw traces carry baseline wander (drift below ~0.5 Hz) on an arbitrary amplitude scale. We remove it with a 4th-order zero-phase 0.5 Hz high-pass [16, 36] and z-score each lead. Normalization is per lead because median amplitude varies ~2.2× across leads; a global scale would let the largest leads dominate the shared backbone. We keep a plain mean/std z-score rather than a robust one, since the within-lead distribution is heavy-tailed (excess kurtosis 15.4) by design: that tail is the QRS complex we must preserve.

The spectrum, estimated with Welch's method [31], fixes the remaining choices. Power at 60 Hz is only 0.62× its neighbours, so the notch is defensive rather than corrective, and ~99% of energy lies below 40 Hz, far under the 125 Hz Nyquist of a 250 Hz signal [28].

The cohort arrives at two native sampling rates: 1,780 recordings at 500 Hz (5,000 samples over the 10 s strip) and 398 at 250 Hz (2,500 samples). Because the high-pass and notch are specified in Hz, we first resample the 2,500-sample records up to 5,000 so the filters act identically on every recording, apply them, then decimate every record by two to a uniform 250 Hz (2,500 samples) for the model input. We standardize *down* to 250 Hz rather than *up* to 500 Hz for three reasons. The 125 Hz Nyquist of 250 Hz already sits far above the ~40 Hz band where the energy concentrates, so no diagnostic content is lost. The 398 lower-rate recordings are natively 250 Hz, so upsampling the whole cohort would fabricate high-frequency detail they never captured. And halving the length to 2,500 samples halves the model's input, parameters, and compute.

3 Exploratory Data Analysis

3.1 Inter-Lead Structure

A *channel-dependent* model mixes the twelve leads from its first layer (a 2D convolution or attention block); a *channel-independent* model runs each lead through the same network separately and combines them only at the end. Mixing earns its extra parameters only if the *relationship between leads* carries class information.

In plain terms, we ask whether preeclampsia changes how the leads relate to one another. We measure each lead pair's tendency to move together (their correlation). We then average those pairwise relationships separately over preeclamptic and normal recordings and ask how much the two averages differ. They barely differ: the inter-lead relationships are almost identical in the two classes. This is a statement about the *relationships*, not the leads themselves: the leads are not all alike, but any given pair co-varies in much the same way whether or not the patient is preeclamptic. The rest of

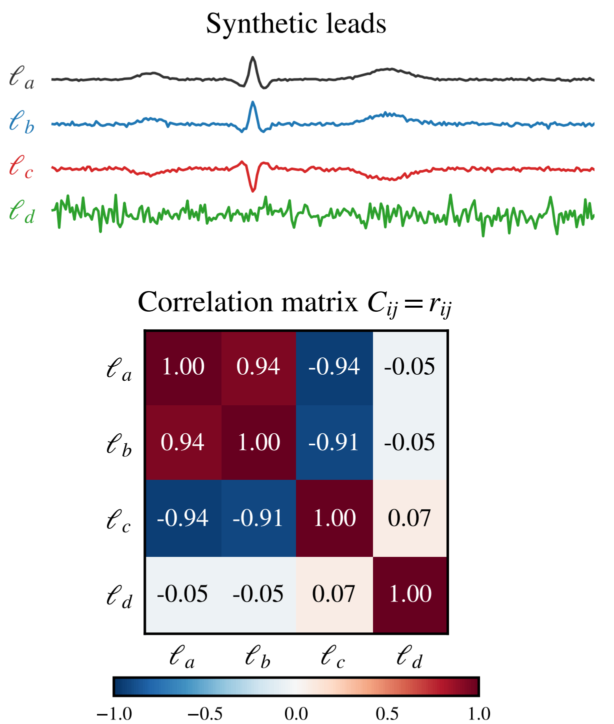

Figure 1: The pairwise operation on four synthetic leads. ℓ_b tracks ℓ_a ($r = 0.94$), ℓ_c is its inverse ($r = -0.94$), and ℓ_d is unrelated ($r = -0.05$); these values populate the correlation matrix C (Eqs. (1)–(2)), which has a unit diagonal.

this subsection makes that measurement precise, building from a single lead pair up to one number.

Two leads. For leads a and b sampled at $t = 1, \ldots, T$, their Pearson correlation measures whether they move alike:

$$r_{ab} = \frac{\sum_t (a_t - \bar{a})(b_t - \bar{b})}{\sqrt{\sum_t (a_t - \bar{a})^2}\,\sqrt{\sum_t (b_t - \bar{b})^2}} \in [-1, 1], \tag{1}$$

with $r_{ab} = 1$ in perfect lockstep, -1 mirrored, and 0 unrelated. The numerator is the covariance of the two leads, while the denominator divides out each lead's own spread, so r_{ab} captures agreement in *shape* regardless of amplitude: a lead and a scaled copy of it score 1.

All pairs of one recording. Stacking r over every lead pair gives that recording's 12×12 correlation matrix, with unit diagonal:

$$C_{ij}^{(n)} = r_{ij}, \qquad i, j \in \{1, \ldots, 12\}, \quad C_{ii}^{(n)} = 1. \tag{2}$$

The matrix is symmetric ($C_{ij}^{(n)} = C_{ji}^{(n)}$), so its content is the 66 distinct off-diagonal correlations. Figure 1 shows these two steps on four synthetic leads.

Each class. Averaging over recordings within a class gives its typical coupling:

$$\bar{C}_{\mathrm{PE}} = \frac{1}{N_{\mathrm{PE}}} \sum_{n \in \mathrm{PE}} C^{(n)}, \qquad \bar{C}_{\mathrm{N}} = \frac{1}{N_{\mathrm{N}}} \sum_{n \in \mathrm{N}} C^{(n)}. \tag{3}$$

Here N_{PE} and N_{N} count the recordings in each class; averaging cancels recording-specific noise and leaves the coupling pattern typical of that class.

Disease effect, per pair. Their entry-wise difference isolates how each lead pair's coupling shifts with disease:

$$D = \bar{C}_{\mathrm{PE}} - \bar{C}_{\mathrm{N}}, \qquad D_{ij} = \text{change in lead } i\text{–}j \text{ coupling}. \tag{4}$$

Since both class matrices have a unit diagonal, D has a zero diagonal: a positive off-diagonal entry marks a lead pair that grows *more* correlated under preeclampsia, a negative one a pair that decouples.

One number. The Frobenius norm (the Euclidean length of the matrix read as a flat list of entries) collapses all of D into a single nonnegative magnitude [14]:

$$\|D\|_F = \sqrt{\sum_{i=1}^{12} \sum_{j=1}^{12} D_{ij}^2} = 0.979. \tag{5}$$

Equations (4)–(5) make the conclusion concrete: each D_{ij} is a verdict on whether one lead pair couples the same way in both classes, and the norm sums those per-pair verdicts into one score. It is 0 only if *every* pair is identical. Here it is 0.979: the off-diagonal entries average just 0.073 and never exceed 0.174 on the $[-1, 1]$ scale, and the two class matrices are indistinguishable by eye (Fig. 2). The leads co-vary almost identically with or without preeclampsia.

A channel-dependent model would therefore spend parameters on cross-lead structure that barely differs between classes; the discriminative signal must instead live *within* each lead. We process every lead independently through one shared-weight 1D backbone and fuse only at the final linear layer (Section 4). A per-lead RMS test confirms no lead carries the signal alone (only 1/12 clears $p < 0.05$), so shared weights, not lead-specific pathways, are the right choice.

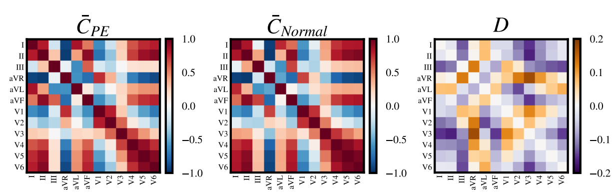

Figure 2: Class-average inter-lead correlation matrices for preeclamptic ($\bar{C}_{\mathrm{PE}}$) and normal ($\bar{C}_{\mathrm{N}}$) recordings and their difference D on the real cohort. The two class matrices are nearly identical and D is near zero everywhere (note the ± 0.2 scale), giving $\|D\|_F = 0.979$.

3.2 Temporal Scale and Alignment

The shape of the per-lead backbone (where in time to look, and how widely) follows from two questions about the time axis.

Is the signal tied to a fixed position? A classifier that flattens the time axis treats absolute position as meaningful; one that pools over time does not. We test which is appropriate by comparing the classes sample-by-sample on a common clock. The recordings are not R-peak aligned, so averaging washes out beat morphology: the class-mean waveform of lead II hovers near zero for both classes,

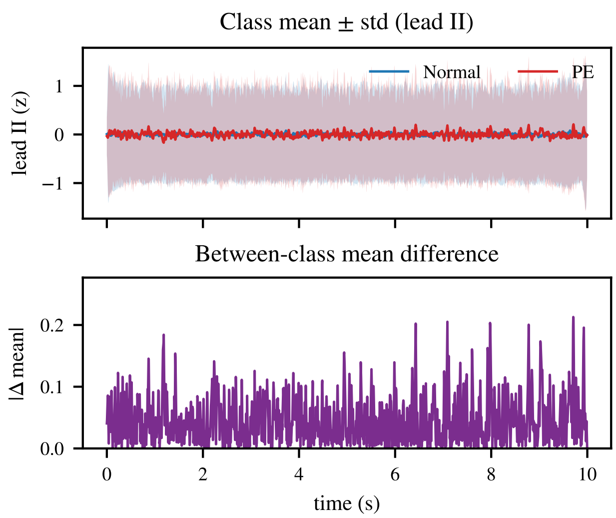

Figure 3: Lead II on a common clock. Top: the class-mean waveforms (PE red, Normal blue) are washed out and their ±std bands overlap almost entirely, so within-class variation dominates. Bottom: the between-class mean difference stays small and is spread across the whole strip, with no localized peak: the signal is not tied to a fixed position.

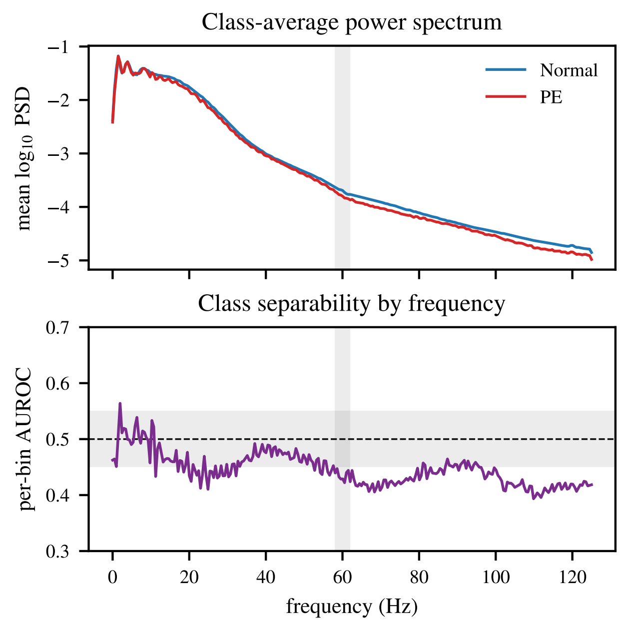

Figure 4: The classes do not separate in the frequency domain; spectra are shown *before* the 60 Hz notch. Top: the class-average power spectra (PE red, Normal blue) are nearly identical, with no prominent mains peak at 60 Hz (shaded). Bottom: the AUROC of each frequency bin's power for separating the classes stays within the chance band (0.45–0.55) across the spectrum.

while the within-class standard deviation is an order of magnitude larger and the two ±std bands overlap almost completely (Fig. 3, top). The between-class mean difference is correspondingly small (at most 0.21 against a within-class std of ≈ 1.0) and *smeared across the entire ten-second strip* rather than concentrated at any latency (Fig. 3, bottom). The discriminative cue is therefore *local waveform morphology wherever it occurs*, not its absolute position. This calls for convolutions, which detect a pattern regardless of where it sits, followed by *global average pooling*, which collapses the time axis and makes the model invariant to where in the strip a beat falls.

At what temporal scale? The morphology that carries the signal is the PQRST complex of a single beat. At 60–100 beats per minute a cardiac cycle lasts 600–1000 ms, and the diagnostic waveform (the P wave, QRS complex, ST segment, and T wave) occupies roughly its first 400 ms [30]. This fixes a requirement on the backbone: each per-lead feature should integrate about one complete PQRST complex, so that a repolarization change can be read relative to the QRS that precedes it. The markers implicated in preeclampsia (QT, T_{pe}, and ST shifts) [4, 25] are all defined this way. Section 4.1.1 shows that the chosen kernels meet this ~one-cycle target with a 444 ms receptive field, after which global average pooling aggregates the per-beat summaries across the whole strip.

3.3 Spectral Separability

A natural alternative to learning from morphology is to classify from the signal's frequency content. We checked whether the classes separate in the power spectrum by estimating each recording's Welch periodogram (averaged over leads) and asking, at each frequency, how well a single bin's power distinguishes the two classes. They do not (Fig. 4). Shown before the 60 Hz notch, the class-average spectra are nearly identical and carry no prominent mains peak, and per-bin separability hovers at chance across the spectrum; the best-performing range (1–5 Hz) reaches only an AUROC of 0.54. This is expected: preeclampsia alters waveform *morphology and relative timing* (T-wave shape, intervals defined against the QRS) [25], which a power spectrum discards along with phase. The result confirms that the discriminative signal is morphological, supporting a model that convolves over the waveform rather than one built on spectral features.

4 Methods

RepNet comprises two models that approach preeclampsia identification from opposite ends of the feature-engineering spectrum. They share only the cohort (Section 2) and the patient-grouped cross-validation used to evaluate them. A recording is $x \in \mathbb{R}^{12\times2500}$ (twelve leads, 2,500 samples at 250 Hz) with a binary label, and both estimate $\mathbb{P}(\text{preeclampsia} \mid x)$; they differ in what they look at.

RepNet-CNN consumes the raw waveform and learns its own features through the shared per-lead backbone motivated in Section 3. RepNet-RF is an entirely separate model that does not use the raw waveform at all; it classifies a vector of *manually derived* ECG features. Using NeuroKit2 [22], it computes a fixed set of clinically defined measurements (QT dispersion, the T_{pe} interval and its rate-corrected ratios, ST-segment elevation and depression, P-wave dispersion, and the frontal QRS–T axis) chosen with our clinical advisors to capture repolarization markers the vendor pipeline omits. It then combines these with the 656 measurements the GE MUSE system already reports for each recording. A decision-tree ensemble classifies the resulting vector: a random forest [7] voted with logistic-regression and support-vector-machine [13] classifiers. Built from different representations of the same signal, the two models also serve as a cross-check: where a learned model and a feature-based one agree, the underlying signal is more credible than either alone would establish.

4.1 RepNet-CNN

4.1.1 Architecture. RepNet-CNN realizes the two findings of Section 3: it processes each lead independently through one shared-weight backbone (Section 3.1), sized to see a single heartbeat (Section 3.2). The input is reshaped from $(12, 2500)$ to $(12, 1, 2500)$ so the same backbone is applied to every lead. That backbone is three stages of Conv1d $\rightarrow$ BatchNorm $\rightarrow$ Mish (Fig. 6) with channel widths $16 \rightarrow 32 \rightarrow 48$, kernels 31, 21, 11, and stride 2 throughout. An adaptive average pool collapses each lead to a 48-dimensional vector; the twelve vectors are concatenated into a 576-dimensional representation and mapped to logits by a single Dropout(0.15), Linear$(576, 2)$ head. Because the backbone weights are shared across all twelve leads, the model totals only 29,490 parameters; the full network is diagrammed in Appendix A (Fig. 5).

Receptive field. The backbone must *see* a full PQRST complex to meet the ~one-cycle requirement of Section 3.2. For a stack of 1D convolutions the receptive field follows the recurrence

$$\mathrm{RF}_\ell = \mathrm{RF}_{\ell-1} + (k_\ell - 1)\, J_{\ell-1}, \qquad J_\ell = \prod_{j=1}^{\ell} s_j, \tag{6}$$

where k_ℓ and s_ℓ are the kernel size and stride of layer ℓ, J_ℓ is the cumulative stride (the *jump* between adjacent outputs), and $\mathrm{RF}_0 = J_0 = 1$. Each stride-2 layer doubles the jump, so the $(k_\ell - 1)\, J_{\ell-1}$ term grows geometrically and a few layers span a wide field cheaply: the same 111-sample reach would need a single 111-tap kernel versus the 31+21+11 = 63 taps used here. The field builds up as in Table 3, reaching 111 samples (444 ms at 250 Hz) by the third stage, about one cardiac cycle. This bounds only the convolutional view; the adaptive average pool then spans the full 2,500-sample (10 s) input, aggregating per-beat summaries over all ten-to-sixteen beats before the leads are fused.

Table 3: Receptive-field buildup of the per-lead backbone, applying Eq. (6) **layer by layer.**

Layer	k_ℓ	s_ℓ	$J_{\ell-1}$	RF_ℓ (samp.)	(ms)
Input	–	–	1	1	4
Conv 1	31	2	1	31	124
Conv 2	21	2	2	71	284
Conv 3	11	2	4	111	444

4.1.2 Training. A fresh model is trained for each split. The objective is a focal cross-entropy [19] with focusing parameter $\gamma = 1$ and label smoothing 0.05,

$$\mathcal{L} = (1 - p_t)^\gamma\, \mathrm{CE}(y, \hat{y}), \tag{7}$$

where p_t is the model's probability for the true class; the $(1-p_t)^\gamma$ factor down-weights already-confident examples. We apply no explicit class weighting: the 15.4% positive rate is handled instead by this down-weighting together with mixup [34] and heavy augmentation, which enlarge the effective positive set rather than re-balance the loss. With probability 0.5 each batch is mixed up: two examples and their labels blended linearly with $\lambda \sim \mathrm{Beta}(0.2, 0.2)$.

We optimize with AdamW [21] (learning rate 1.2×10^{-3}, weight decay 5×10^{-3}) under a cosine schedule annealing to 10^{-6} over the 80-epoch budget, with batch size 64 and two-step gradient accumulation (effective batch 128). Training stops early once the validation AUROC has not improved for 20 epochs, and the best-AUROC weights are restored. Validation uses the inner fold of the patient-grouped split (Section 2); no test data is seen during training, and a single model is trained per split (no bagging).

Learning-rate schedule. The learning rate is not held fixed but annealed with a cosine schedule [20]. Indexing epochs by $t = 0, 1, \ldots, T_{\max}$, the rate at epoch t is

$$\eta_t = \eta_{\min} + \tfrac{1}{2}\,(\eta_{\max} - \eta_{\min})\left(1 + \cos\frac{\pi t}{T_{\max}}\right), \tag{8}$$

which traces one half-period of a cosine from the initial rate $\eta_{\max} = 1.2 \times 10^{-3}$ at $t = 0$ down to the floor $\eta_{\min} = 10^{-6}$ at $t = T_{\max}$. We set $T_{\max} = 80$ epochs (matching the epoch budget) and step the schedule once per epoch (`CosineAnnealingLR` in the released code). The cosine shape gives a deliberately uneven decay: the rate is nearly flat early on, so the model takes large, exploratory steps while the loss surface is far from any minimum; it falls fastest through the middle epochs; then it flattens again near the floor, so late updates are small and the weights settle into a minimum rather than bouncing around it. Compared with a fixed rate this trades early speed for late stability without manual step-decay tuning, and compared with a linear decay the flat tails spend more budget at the two rates that matter most: high for exploration, low for convergence. In practice early stopping (patience 20) usually halts a split before epoch 80, so the rate typically reaches the lower-middle of the curve rather than the 10^{-6} floor; $T_{\max}$ therefore sets the *shape* of the decay, not a guaranteed endpoint. We use a single cosine run with no warm restarts.

Activation. Each backbone stage uses the Mish activation [23] in place of the usual ReLU,

$$\mathrm{Mish}(x) = x\, \tanh\!\big(\mathrm{softplus}(x)\big) = x\, \tanh\!\big(\ln(1 + e^x)\big), \tag{9}$$

the self-gated form already shown per block in Fig. 6. Two properties motivate the choice over ReLU, both bearing on how gradients flow during training. First, Mish is smooth and continuously differentiable everywhere, whereas ReLU has a non-differentiable kink at the origin and a flat zero branch; the smoother response gives a better-conditioned loss surface and non-vanishing gradients through the negative regime. Second, Mish is non-monotonic and bounded below (it dips to ≈ -0.31 before rising), so a unit driven slightly negative still passes a small signal and a non-zero gradient. ReLU instead zeros every negative pre-activation and its gradient there, so units that drift negative can stop updating entirely, the "dying ReLU" failure mode. This matters most for a model this small: with only 29,490 parameters there is little spare capacity to lose to dead units. Mish was introduced as a self-regularizing, non-monotonic activation and reported to improve accuracy over ReLU and Swish across a range of vision benchmarks [23]; we adopt it for these properties rather than from an in-house activation ablation.

Every training waveform is augmented on the fly with seven stochastic, label-preserving transforms (Table 4), each mimicking a plausible source of recording variation. Together they expand the effective training set, valuable given only ∼230 positive recordings per training fold.

Table 4: The seven on-the-fly training augmentations, applied to each 250 Hz waveform with probability p.

Augmentation	p	Parameters
Gaussian noise	0.50	$\sigma \in [0.01, 0.05]$
Amplitude scale (per lead)	0.50	±15%
Time shift (circular)	0.30	±150 samples
Lead dropout	0.10	each lead zeroed w.p. 0.12
Cutout	0.25	zero a 50–200-sample window
Baseline wander	0.20	sine, 0.05–0.5 Hz, amplitude ≤ 0.2
Resample / time-warp	0.10	rate 1 ± 0.05

4.2 RepNet-RF

4.2.1 Feature Engineering. RepNet-RF classifies hand-engineered ECG features rather than the raw waveform. Each of the 2,191 recordings that has a waveform arrives with 656 measurements already computed by the GE MUSE system, which we call the *metadata*; with guidance from our clinical advisor Dr. Lihong Mo, we additionally extract features linked to preeclampsia that MUSE does not report. NeuroKit2 feature extraction fails on a small number of recordings (for example when delineation returns missing landmarks), and those are dropped; to keep the comparison fair, the MUSE-derived features drop the same recordings. Table 5 gives the resulting counts for the full (unbalanced) cohort and for a class-balanced subset used in some experiments.

Table 5: Recordings used by RepNet-RF, before and after dropping cases where NeuroKit2 extraction failed (preeclamptic / normal).

Dataset	Original	Dropped	Final
Unbalanced	337 / 1854	8 / 24	329 / 1830
Balanced	187 / 182	6 / 1	181 / 181

NeuroKit2 provides routines for cleaning ECG signals and locating fiducial landmarks. Because raw traces drift from a common baseline, a baseline-wander filter is applied before delineation, after which we compute the features in Table 10. Our graduate mentor Hana Shaik had previously implemented the ST–T abnormality features for detecting regional wall motion abnormalities [17]; we validated our implementation against hers and on the Nightingale ECG dataset, confirming the clinical interpretation of each formula.

The frontal QRS and T axes were the hardest to compute. We implement the QRS axis two ways: an isoelectric method that locates the isoelectric lead and reads ±90° from the sign of the perpendicular lead, and a direct arctan formula that returns a specific angle [1, 3], though in clinical practice small angular differences are insignificant [3]. The T axis uses the analogous arctan form [26], and the frontal QRS–T angle is their absolute difference. Dr. Ebong, a UC Davis Health cardiologist, is evaluating the accuracy of these implementations.

4.2.2 Comparison with MUSE Features. To check the NeuroKit features, we recomputed the same quantities from the MUSE metadata (Table 11) on the identical set of recordings. Averaged across recordings, the NeuroKit values are consistently lower than the MUSE values. We briefly trialed an alternative library, ECGDataKit, whose functions more closely mirror MUSE: it gave a closer median R-peak amplitude but failed on the 250 Hz recordings and offered fewer feature routines, so we retained NeuroKit2.

The NeuroKit2 and MUSE feature definitions are listed in Appendix B.

4.2.3 Classifier. Each feature set is classified by a random forest inside a stratified group k-fold split keyed on the obfuscated patient MRN, which prevents leakage, with a random undersampler applied within the training folds [6, 18]. We evaluate three feature sources (the NeuroKit2 features, their MUSE-reported counterparts, and either combined with the full MUSE metadata) and report their performance in Section 5.2.

4.3 Evaluation Protocol

Both models are evaluated with *patient-grouped, stratified* cross-validation built on scikit-learn's `StratifiedGroupKFold`. Grouping on the obfuscated patient MRN guarantees that no patient appears in both a training and a test partition: without it, the 428 patients with multiple recordings (Section 2) would leak identity across the split and inflate every metric. Stratification holds the 15.4% preeclampsia prevalence fixed in every fold, so each test set reflects the clinical base rate. The two models differ only in how the folds are used.

RepNet-RF. A single five-fold `StratifiedGroupKFold` (`shuffle=True`, fixed random seed): each of the five folds serves once as the 20% held-out test set while the model trains on the other four, with a `RandomUnderSampler` balancing the training folds. The reported metrics are the mean over the five test folds.

RepNet-CNN. Rather than a single five-fold pass, the CNN is evaluated over *thirty independent* patient-grouped 80/20 splits, each seeded by $s_i = 7i + 1000$ for split $i = 0, \ldots, 29$. Within a split, an outer `StratifiedGroupKFold` (five folds, `random_state` $= s_i$) supplies one fold as the 20% held-out test set and the remaining 80% as a development set; an inner `StratifiedGroupKFold` (eight folds, `random_state` $= s_i + 1$) on the development set carves off a validation fold for early stopping (Section 4.1.2). Each split trains a fresh model, and we report the mean ± standard deviation of every metric across the thirty splits. The randomized seeding turns evaluation into a distribution rather than a single fold: the spread directly estimates run-to-run stability on this small, imbalanced cohort, and the best and median splits supply the released checkpoints.

For both models we report threshold-free discrimination (AUROC and AUPRC, the latter against the 0.154 prevalence baseline) together with two operating points: Youden's J [33], which maximizes sensitivity + specificity −1, and a screening point that fixes sensitivity at ≥ 80% and reports the resulting specificity and negative predictive value.

5 Results

5.1 RepNet-CNN

RepNet-CNN attains a mean AUROC of 0.709 ± 0.049 over the 30 patient-grouped splits (Table 6). Scores range from 0.594 on the hardest split to 0.779 on the easiest, a spread that justifies reporting the distribution rather than a single fold. Its AUPRC of 0.342 ± 0.075 sits well above the 0.154 no-skill baseline set by the prevalence. The model works best as a rule-out screen. At a threshold fixed to 80% sensitivity it holds a negative predictive value of 0.928 ± 0.019, its most stable metric across splits, so a negative prediction meaningfully lowers the probability of preeclampsia. Its specificity (0.50) and precision are more modest, so a positive prediction warrants confirmatory assessment rather than standing alone.

Table 6: RepNet-CNN performance by operating point over 30 patient-grouped splits (percent, mean ± std). AUROC and AUPRC are threshold-free.

Operating point	AUROC	AUPRC	Spec.	Sens.	Bal. Acc.	F1
Youden's J (τ=0.61)	70.9±4.9	34.2±7.5	62.7±12.6	73.0±9.2	67.9±3.8	40.4±6.9
Sens. ≥ 80% (τ=0.55)	70.9±4.9	34.2±7.5	49.6±8.9	**80.7±0.5**	65.1±4.4	36.7±6.1

The ROC and precision–recall curves on the best split (AUROC 0.779, average precision 0.485 against the 0.16 prevalence baseline) are shown in Appendix A (Fig. 7). To see *where* the network looks, we attribute the preeclampsia logit back to the Lead II waveform with Integrated Gradients [29] and Grad-CAM [27] (Fig. 8). On a confident true positive the attribution toward preeclampsia concentrates on the QRS complexes and ST segments, and it reverses sign on a true negative; the network keys on clinically meaningful morphology rather than spurious artefacts.

5.2 RepNet-RF

Despite the systematic offset between the libraries, the NeuroKit- and MUSE-derived features perform comparably (Table 7): NeuroKit under-reports values but does so consistently, so the classifier is largely unaffected. Adding the extracted features to the full MUSE metadata expands the set to 721 features (723 with NeuroKit) and improves performance (Table 8), although a model trained on the metadata alone scored slightly higher, a hint that the enlarged feature set begins to overfit.

Table 7: Random-forest performance with NeuroKit2- vs MUSE-derived features (percent).

Features	AUROC	AUPRC	Spec.	Sens.	Acc.	Prec.	F1
Balanced, NeuroKit	66.22	67.53	66.26	57.49	61.87	62.99	59.98
Balanced, MUSE	62.47	65.66	60.27	59.73	59.95	60.22	59.75
Unbalanced, NeuroKit	63.19	22.62	63.83	54.73	62.44	21.33	30.67
Unbalanced, MUSE	64.25	24.68	64.75	55.33	63.32	22.03	31.44

Table 8: Performance when the extracted features are added to the full MUSE metadata (percent).

Features	AUROC	AUPRC	Spec.	Sens.	Acc.	Prec.	F1
MUSE + NeuroKit, bal.	70.37	72.94	67.92	63.89	65.85	67.57	65.39
MUSE + MUSE-ext., bal.	72.03	73.90	70.10	62.78	66.38	68.21	65.27
MUSE + NeuroKit, unbal.	69.93	32.55	73.74	56.12	71.03	27.97	37.20
MUSE + MUSE-ext., unbal.	69.78	34.12	73.19	57.32	70.74	28.18	37.66

A single random forest tops out near 72% AUROC (Tables 7 and 8); combining it with a support-vector machine and a logistic-regression classifier lifts performance further (Table 9). The strongest RepNet-RF configuration (an unweighted vote of the random forest, SVM, and logistic regression) reaches a mean AUROC of 75.7 ± 3.9% over the five folds, with 62.4% sensitivity and 74.6% specificity at its operating point. Weighted voting, stacking, and a bagged CatBoost ensemble all land within a percentage point, so the gain comes from model averaging rather than any single ensemble recipe.

Table 9: RepNet-RF ensemble performance (percent, mean ± std over the five patient-grouped folds). The best row is an unweighted vote of the random forest with an SVM and logistic regression.

Ensemble	AUROC	AUPRC	Spec.	Sens.	Bal. Acc.	F1
Unweighted voting	74.54±4.16	41.81±7.86	74.26±3.95	59.32±10.70	66.79±3.60	38.94±2.99
Weighted voting	74.75±4.35	42.26±7.92	74.75±4.48	60.23±10.88	67.49±3.45	39.80±2.76
Unwtd. + SVM/LR	**75.72±3.94**	41.58±8.03	74.64±4.70	**62.35±10.93**	**68.50±3.22**	**40.85±2.24**
Stacking	74.77±4.35	42.43±7.59	74.37±4.18	59.62±10.80	67.00±3.46	39.17±2.79
Bagging CatBoost	74.81±4.98	41.96±8.86	74.70±4.68	59.62±15.19	67.16±5.59	39.10±5.21

An importance analysis of the engineered features identifies which leads and which measurements drive the classification (Appendix B). The inferior and anterior precordial leads (III and V1–V3) contribute the most important features (Fig. 9), and the most frequently selected measurements are ST-shift flags, the QRS intrinsicoid deflection, the ST-to-T-peak interval, and P- and T-wave durations (Fig. 10), depolarization- and repolarization-timing features consistent with the preeclampsia literature [5, 25].

6 Discussion

6.1 Interpretation

The two models finish close (AUROC 0.709 for RepNet-CNN against 0.757 for RepNet-RF), and the small gap is itself the finding. RepNet-RF wins because this is exactly the regime where a feature-based model should: a restricted data domain with strong, known features. Its inputs already encode how preeclampsia deforms the

ECG, so it does not have to discover that structure, only weight it. RepNet-CNN gets no such prior; it learns the same repolarization and morphology features from raw waveforms, and with only 335 positive recordings it is data-limited: there are too few examples for a network to recover from scratch what the engineered features hand the forest outright. The neural network is not worse in principle; it is starved, and the gap is the price of learning features the random forest is simply given.

6.2 Limitations

Our results come with real caveats. The cohort is single-site (2,178 ECGs from one health system), so we cannot yet claim the models generalize to other populations, machines, or acquisition settings; Butler et al. validated on an external cohort, and we have not [8]. It is also small and imbalanced (335 positives, 15.4%), which both caps the ceiling (our AUROCs sit well below Butler's 0.85 [8], plausibly a dataset-size effect) and widens the per-split spread we report.

The models themselves leave room. RepNet-RF runs on default hyperparameters behind only a chi-squared feature filter, so its 0.757 is a floor rather than a tuned ceiling. RepNet-CNN's raw scores are poorly calibrated (Brier 0.48 on the best split), so they have to be thresholded rather than read as probabilities. And the two pipelines clean the data slightly differently (2,178 recordings for the CNN against 2,191 before the RF's own drops), a small bookkeeping gap we have not reconciled.

6.3 Future Work

The most immediate extension is lead reduction. RepNet-RF already tells us which leads matter: its importance analysis (Appendix B) concentrates the discriminative features in a handful of leads, so we can prune the rest. A pilot ranking features with CatBoost on the smaller balanced cohort recovered comparable accuracy from only six leads; the simpler model gave up some balanced accuracy, but most of the preeclampsia signal clearly survives aggressive lead reduction, echoing Xue's result for general morphology [32]. Fewer leads means a cheaper, more wearable-friendly test, the direction Butler et al. point toward with their single-lead work [10].

A larger, multi-site dataset would open two further directions. First, temporal analysis: with enough longitudinal recordings we could measure how far in advance of clinical onset the models detect preeclampsia, as Butler et al. do [8], turning identification into early warning. Second, a hybrid model that fuses the waveform RepNet-CNN learns with the engineered features RepNet-RF relies on, which we expect to push balanced accuracy past either model alone.

7 Conclusion

We set out to identify preeclampsia directly from the 12-lead ECG, and both of our models do. RepNet-CNN, a 29,490-parameter channel-independent network, reaches a mean AUROC of 0.709 ± 0.049 on raw waveforms; RepNet-RF, an ensemble over interpretable NeuroKit2 and MUSE features, reaches 0.757 ± 0.039. Neither is a deployable screen yet (their operating points still trade away the specificity a real test needs), but both clear the harder bar of carrying genuine, patient-grouped signal, and their explanations land on the same clinically established markers: QRS morphology, ST-segment shifts, and the T_{pe} interval [25, 30]. Together they show that the preeclampsia ECG waveform carries genuine signal, pointing toward a more efficient and effective form of diagnosis and patient care.

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A RepNet-CNN

This appendix collects the architecture and diagnostics behind RepNet-CNN. Figure 5 diagrams the full network — twelve leads passed through one shared per-lead backbone, average-pooled, concatenated, and fused by a single linear head — while Fig. 6 expands the operations inside one backbone block. Figure 7 then reports the ROC and precision–recall curves on the best split, and Fig. 8 shows *where* the network looks, attributing the preeclampsia logit back to the Lead II waveform.

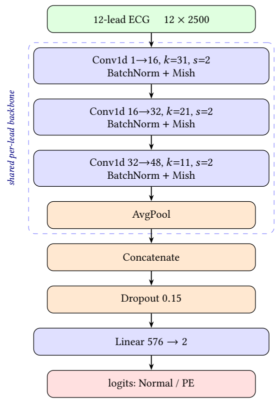

Figure 5: RepNet-CNN architecture (29,490 parameters). The 12×2500 input is reshaped to $(12, 1, 2500)$ so the dashed shared per-lead backbone — three Conv1d→BatchNorm→Mish stages with channel widths $16 \to 32 \to 48$ and kernels $31, 21, 11$ at stride 2 — is applied identically to every lead. Average pooling collapses each lead to a 48-d vector; the twelve are concatenated into a 576-d representation and mapped to Normal/PE logits by a single $\mathrm{Dropout}(0.15)$, $\mathrm{Linear}(576, 2)$ head. Sharing the backbone across all twelve leads keeps the parameter count at $29{,}490$.

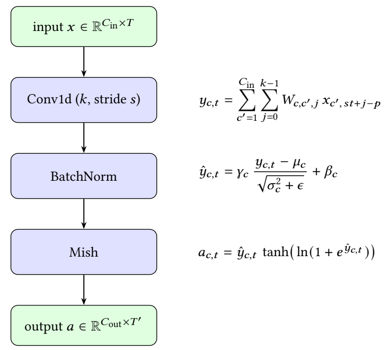

Figure 6: One backbone block, unrolled with its governing equations (right): a bias-free Conv1d of kernel k and stride s, BatchNorm [15] rescaling by learned γ_c, β_c over batch statistics μ_c, σ_c^2, and the Mish activation. The three stages instantiate this block with $(k, C_{\text{out}}) = (31, 16), (21, 32), (11, 48)$ at $s = 2$ throughout, the same backbone applied to every lead in Fig. 5.

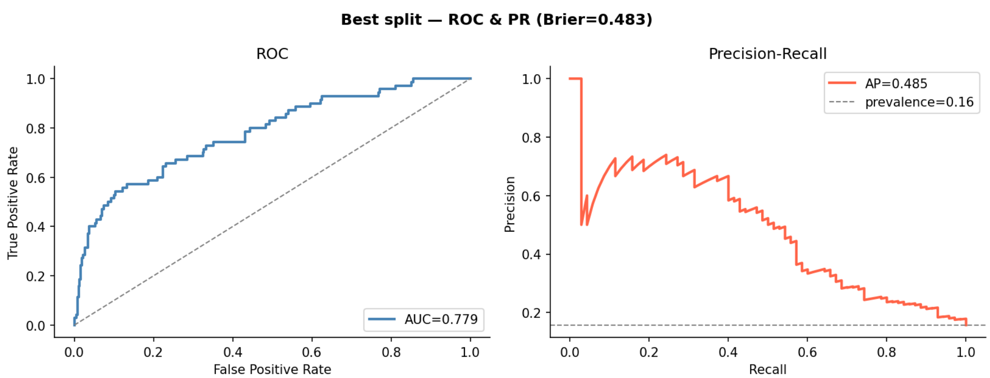

Figure 7: Discrimination on the best of the 30 patient-grouped splits (AUROC 0.779, average precision 0.485). Left: the ROC curve. Right: the precision–recall curve, whose 0.16 baseline is the no-skill rate set by preeclampsia prevalence. This split is the upper end of the 0.709 ± 0.049 AUROC distribution reported in Table 6.

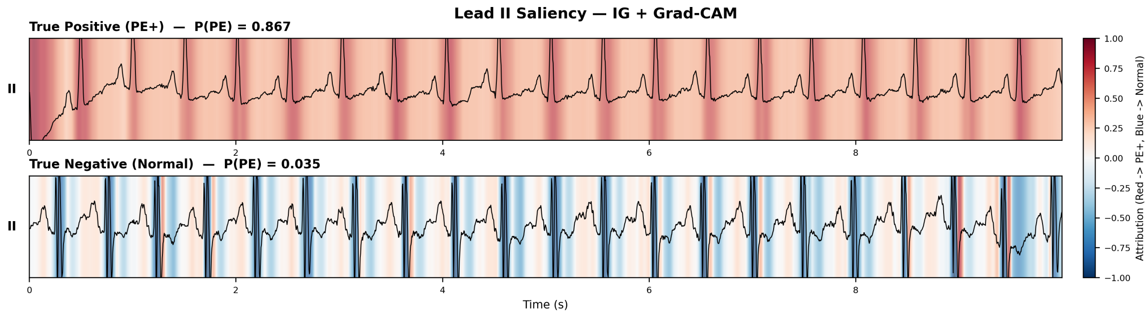

Figure 8: Lead II saliency from attributing the preeclampsia logit back to the waveform with Integrated Gradients and Grad-CAM: red marks samples pushing the prediction toward preeclampsia, blue toward normal. On a confident true positive (top) the attribution concentrates on the QRS complexes and ST segments; it reverses sign on a true negative (bottom), indicating the network keys on clinically meaningful morphology rather than spurious artefacts.

B RepNet-RF Features and Lead Importance

The features RepNet-RF computes with NeuroKit2 [22] are defined in Table 10, and the same quantities derived from the GE MUSE-reported fields in Table 11. These are standard electrocardiographic constructs [30], with the repolarization-dispersion measures (T_{pe}, QT dispersion, and their rate-corrected ratios) following established clinical definitions [4, 25]. Figure 9 then counts the important engineered features per lead, and Fig. 10 how often each measurement is flagged as important across the twelve leads.

Table 10: ECG features extracted with NeuroKit2. $P_{\mathrm{on}}/P_{\mathrm{off}}$ are P-wave on/offsets, R_{on} the QRS onset, and T_{off}, T_{peak} the T-wave offset and peak; subscript ℓ ranges over the twelve leads.

Feature	Definition
P_{max}	$\max_\ell(P^\ell_{\mathrm{off}} - P^\ell_{\mathrm{on}})$
P_{min}	$\min_\ell(P^\ell_{\mathrm{off}} - P^\ell_{\mathrm{on}})$
P-wave dispersion	$P_{\mathrm{max}} - P_{\mathrm{min}}$
QT dispersion	$\max_\ell(T^\ell_{\mathrm{off}}-R^\ell_{\mathrm{on}}) - \min_\ell(T^\ell_{\mathrm{off}}-R^\ell_{\mathrm{on}})$
T_{pe} interval	$\mathrm{mean}(T_{\mathrm{off}} - T_{\mathrm{peak}})$
T_{pe}/QT	T_{pe} / $\mathrm{mean}(T_{\mathrm{off}} - R_{\mathrm{on}})$
T_{pe}/QTc	$(T_{pe}/\mathrm{QT})\sqrt{\Delta R_{\mathrm{peak}}}$
ST elevation	$\mathbf{1}[>50\%\text{ elevated}]$
ST depression	$\mathbf{1}[>50\%\text{ depressed}]$
QRS axis	isoelectric or arctan method
T axis	isoelectric or arctan method
Frontal QRS–T angle	$\lvert$ QRS axis − T axis $\rvert$

Table 11: The same features from GE MUSE-reported fields. BITFLGS is the ST-shift bit field; RAxis, TAxis **are the reported axes.**

Feature	Definition
P_{max}	$\max_\ell(P_{\mathrm{Dur}})$
P_{min}	$\min_\ell(P_{\mathrm{Dur}})$
P-wave dispersion	$P_{\mathrm{max}} - P_{\mathrm{min}}$
QT dispersion	$\max_\ell(\mathrm{QT}_\ell) - \min_\ell(\mathrm{QT}_\ell)$
T_{pe} interval	$T_{\mathrm{Dur}} + T'_{\mathrm{Dur}} - \mathrm{STEtoTPeak}$
T_{pe}/QT	T_{pe} interval / QT
T_{pe}/QTc	$(T_{pe}/\mathrm{QT})\sqrt{\mathrm{RR}}$
ST elevation	(BITFLGS & 8) $\neq 0$
ST depression	(BITFLGS & 4) $\neq 0$
QRS axis	RAxis
T axis	TAxis
Frontal QRS–T angle	$\lvert$ RAxis − TAxis $\rvert$

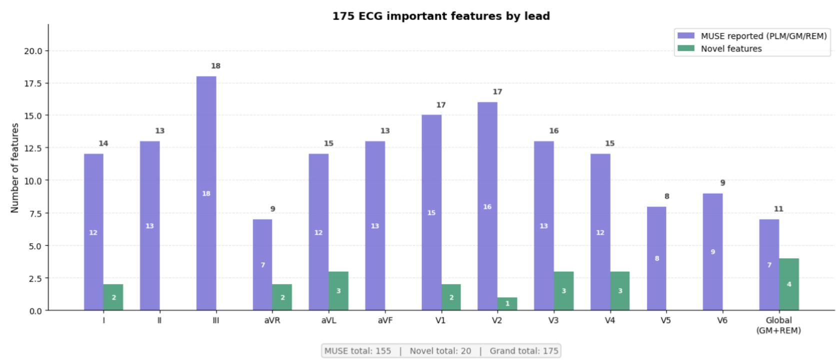

Figure 9: Count of important engineered features per lead for RepNet-RF, split into MUSE-reported and novel (NeuroKit-derived) features.

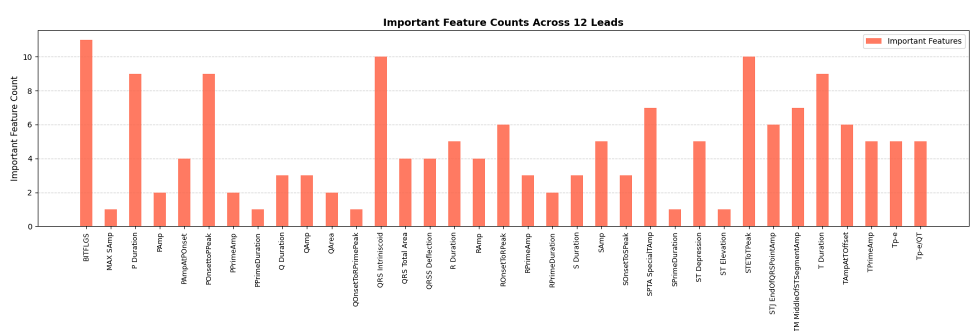

Figure 10: How often each engineered measurement is flagged as important across the twelve leads.